Mini Project Report
on

"SkillBridge: Student Skill Certification Tracker"

Under the Course
"Database Management System Laboratory"

Submitted By

Viraj Pathare	PRN: 124B1F149
Kaustubh Nahire	PRN: 124B1F157
Siddhesh Patil	PRN: 124B1F146
Adi Maitre	PRN: 124B1F151

Under the guidance of
Mrs. Alpana Borse

Department of Information Technology
Pimpri Chinchwad Education Trust's
Pimpri Chinchwad College of Engineering

Academic Year 2025–26

INDEX

Sr. No.	Chapter / Section	Page No.
1	Abstract	2
2	Introduction	3
3	Objectives	3
4	Database Design	4
4.1	ER Diagram	4
4.2	Tables and Attributes	4
4.3	Primary Keys and Foreign Keys	6
5	Implementation	6
5.1	Table Creation (DDL)	6
5.2	Data Insertion	7
5.3	Query Operations	8
6	Results and Output	9
7	Conclusion	10
8	References	10

1. Abstract

SkillBridge is a web-based student skill certification tracker designed to manage and monitor the academic and professional development of engineering students. The system provides a centralized platform where students can maintain profiles, track technical skills, upload certifications, and record participation in workshops and extracurricular activities. It leverages a Supabase backend for efficient database management and real-time operations. The platform also incorporates verification workflows, badge systems, and analytics dashboards to enhance credibility, transparency, and usability. This project demonstrates practical implementation of database design concepts — including relational models, Entity-Relationship diagrams, normalization, and SQL queries — in a real-world application environment. The outcome is a scalable and responsive system that bridges the gap between student achievement and recruiter accessibility.

2. Introduction

In the modern educational ecosystem, students are expected to maintain a strong portfolio of skills, certifications, project experiences, and extracurricular achievements. With increasing competition in the job market, having a well-organized and verifiable record of one's capabilities has become essential. However, most academic institutions lack a centralized and structured system to track such information efficiently. Students often maintain scattered records across multiple platforms — such as LinkedIn, personal resumes, and email attachments — making it cumbersome to present a consolidated view of their accomplishments. This fragmented approach creates challenges not only for students but also for administrators and recruiters who need to assess academic portfolios systematically.

SkillBridge is designed to solve this problem by providing a unified digital platform for managing student profiles, technical skills, certifications, workshop participation, and achievement badges. The system enables students to maintain an up-to-date digital portfolio while allowing administrators and recruiters to verify and analyze the data efficiently. Built using modern technologies such as React for the frontend and Supabase as the backend-as-a-service platform, the application ensures scalability, real-time data synchronization, and ease of use. The project applies core database management concepts including relational schema design, normalization, primary and foreign keys, joins, and SQL queries, making it a highly relevant demonstration of theoretical knowledge applied in a practical software environment.

3. Objectives

- To design a relational database system for SkillBridge: Student Skill Certification Tracker.
- To implement SQL queries using core database management concepts including DDL, DML, and DQL.
- To create a structured and normalized platform for managing student profiles and academic data.
- To enable certification upload, tracking, and verification workflows for authenticity.
- To analyze skill gaps and improve student career readiness through data-driven insights.
- To demonstrate practical use of ER diagrams, primary keys, foreign keys, and relational joins.

4. Database Design

4.1 ER Diagram

The Entity-Relationship (ER) diagram represents the overall structure of the SkillBridge database and defines the relationships between its core entities. The main entities are Student, Skill, Certification, Workshop, Verification, and Faculty. The Student entity is the central component — each student can possess multiple skills (Has Skill), obtain multiple certifications (Obtains), and attend multiple workshops (Attends). Certifications are linked to skills via a skill_id foreign key and go through a Verification process overseen by Faculty members. All relationships are modeled using one-to-many associations with foreign keys ensuring referential integrity throughout the schema.

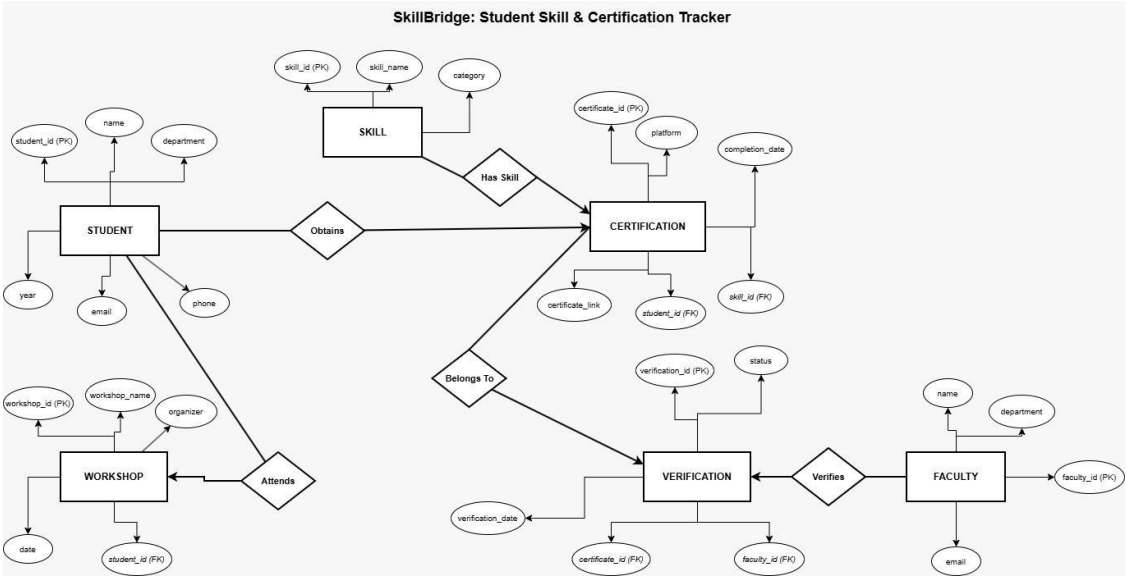

Figure 1: Entity-Relationship (ER) Diagram of SkillBridge Database

4.2 Tables and Attributes

The database consists of five interrelated tables designed to store all relevant data for the SkillBridge platform. Each table is described below along with its attributes and key constraints.

1. Students Table — Stores essential personal and academic information about each registered student.

Attribute	Data Type	Constraint	Description
student_id	SERIAL	PRIMARY KEY	Unique identifier for each student
name	VARCHAR(100)	NOT NULL	Full name of the student
email	VARCHAR(100)	UNIQUE, NOT NULL	Institutional email address
branch	VARCHAR(50)	NOT NULL	Academic branch / department
graduation_year	INT	NOT NULL	Expected year of graduation

2. Skills Table — Records technical and soft skills associated with each student.

Attribute	Data Type	Constraint	Description
skill_id	SERIAL	PRIMARY KEY	Unique identifier for each skill record
student_id	INT	FOREIGN KEY	References students(student_id)
skill_name	VARCHAR(100)	NOT NULL	Name of the skill (e.g., Python, SQL)
proficiency_level	VARCHAR(50)	NOT NULL	Beginner / Intermediate / Advanced

3. Certifications Table — Stores certification details including issuing organization and verification status.

Attribute	Data Type	Constraint	Description
cert_id	SERIAL	PRIMARY KEY	Unique identifier for each certification
student_id	INT	FOREIGN KEY	References students(student_id)
cert_name	VARCHAR(150)	NOT NULL	Name of the certification
issuing_organization	VARCHAR(100)	NOT NULL	Organization that issued the certificate
verification_status	VARCHAR(20)	DEFAULT Pending	Pending / Verified / Rejected

4. Workshops Table — Tracks workshop and seminar participation records for each student.

Attribute	Data Type	Constraint	Description
workshop_id	SERIAL	PRIMARY KEY	Unique identifier for workshop record
student_id	INT	FOREIGN KEY	References students(student_id)
workshop_name	VARCHAR(150)	NOT NULL	Name of the workshop or event
date	DATE	NOT NULL	Date of the workshop

5. Badges Table — Records digital badges and achievement recognitions earned by students.

Attribute	Data Type	Constraint	Description
badge_id	SERIAL	PRIMARY KEY	Unique identifier for each badge
student_id	INT	FOREIGN KEY	References students(student_id)
badge_name	VARCHAR(100)	NOT NULL	Title of the badge earned
description	TEXT	NULLABLE	Details about the badge and its criteria

4.3 Primary Keys and Foreign Keys

Primary keys uniquely identify each record within a table. In the SkillBridge schema, every table uses a SERIAL primary key that auto-increments. Foreign keys establish and enforce relationships between tables — the student_id column appears as a foreign key in the Skills, Certifications, Workshops, and Badges tables, all referencing the primary Students table. This design ensures referential integrity: no record in a child table can reference a student that does not exist in the parent table. ON DELETE CASCADE rules ensure that when a student is removed, all associated records are automatically cleaned up.

5. Implementation

5.1 Table Creation (DDL Queries)

The following SQL statements create the five core tables of the SkillBridge database using PostgreSQL syntax, fully compatible with the Supabase platform. The SERIAL data type provides auto-incrementing primary keys, and REFERENCES enforces foreign key constraints with cascading deletes.

SQL — CREATE TABLE Statements:

```
-- Create Students Table
CREATE TABLE students (
  student_id      SERIAL PRIMARY KEY,
  name            VARCHAR(100) NOT NULL,
  email           VARCHAR(100) UNIQUE NOT NULL,
  branch          VARCHAR(50)  NOT NULL,
  graduation_year INT          NOT NULL
);

-- Create Skills Table
CREATE TABLE skills (
  skill_id          SERIAL PRIMARY KEY,
  student_id        INT REFERENCES students(student_id) ON DELETE CASCADE,
  skill_name        VARCHAR(100) NOT NULL,
  proficiency_level VARCHAR(50)  NOT NULL
);

-- Create Certifications Table
CREATE TABLE certifications (
  cert_id              SERIAL PRIMARY KEY,
  student_id           INT REFERENCES students(student_id) ON DELETE CASCADE,
  cert_name            VARCHAR(150) NOT NULL,
  issuing_organization VARCHAR(100) NOT NULL,
  verification_status  VARCHAR(20)  DEFAULT 'Pending'
);

-- Create Workshops Table
CREATE TABLE workshops (
  workshop_id   SERIAL PRIMARY KEY,
  student_id    INT REFERENCES students(student_id) ON DELETE CASCADE,
  workshop_name VARCHAR(150) NOT NULL,
  date          DATE NOT NULL
);

-- Create Badges Table
CREATE TABLE badges (
  badge_id    SERIAL PRIMARY KEY,
  student_id  INT REFERENCES students(student_id) ON DELETE CASCADE,
  badge_name  VARCHAR(100) NOT NULL,
  description TEXT
```

```
);
```

5.2 Data Insertion (DML Queries)

Sample data is inserted to populate the tables and facilitate testing of all SQL operations. The INSERT statements below demonstrate multi-row insertion syntax supported by PostgreSQL.

SQL — INSERT Statements:

```
-- Insert sample students
INSERT INTO students (name, email, branch, graduation_year) VALUES
  ('Siddhesh Patil',  'siddhesh@pccoe.edu',  'Computer Engineering',   2026),
  ('Kaustubh Nahire', 'kaustubh@pccoe.edu',  'Computer Engineering',   2026),
  ('Viraj Pathare',   'viraj@pccoe.edu',     'Information Technology', 2026),
  ('Adi Maitre',      'adi@pccoe.edu',       'Information Technology', 2026);

-- Insert skills
INSERT INTO skills (student_id, skill_name, proficiency_level) VALUES
  (1, 'Python',           'Advanced'),
  (1, 'SQL',              'Intermediate'),
  (2, 'React',            'Advanced'),
  (3, 'Machine Learning', 'Beginner'),
  (4, 'Figma',            'Intermediate');

-- Insert certifications
INSERT INTO certifications
  (student_id, cert_name, issuing_organization, verification_status)
VALUES
  (1, 'AWS Cloud Practitioner', 'Amazon Web Services', 'Verified'),
  (2, 'Google UX Design',       'Google',              'Pending'),
  (3, 'NPTEL Python',           'IIT Madras',          'Verified');

-- Insert workshops
INSERT INTO workshops (student_id, workshop_name, date) VALUES
  (1, 'Full-Stack Web Dev Bootcamp',    '2025-09-10'),
  (2, 'Supabase & PostgreSQL Workshop', '2025-11-15'),
  (3, 'AI/ML Fundamentals Seminar',     '2025-12-01');

-- Insert badges
INSERT INTO badges (student_id, badge_name, description) VALUES
  (1, 'Top Coder',        'Excellence in competitive programming'),
  (2, 'UI Innovator',     'Outstanding frontend design skills'),
  (3, 'Research Scholar', 'Paper submitted at ISTE 2025');
```

5.3 Query Operations (DQL)

The following SQL queries demonstrate the full range of data retrieval and manipulation operations supported by the SkillBridge database — including joins, aggregations, filtering, sub-queries, updates, and deletions.

SQL — SELECT, JOIN, Aggregate, and Sub-query Operations:

```
-- 1. Retrieve all students
SELECT * FROM students;

-- 2. List students with their skills (INNER JOIN)
SELECT s.name, sk.skill_name, sk.proficiency_level
FROM students s
JOIN skills sk ON s.student_id = sk.student_id;

-- 3. Show verified certifications with student names
SELECT s.name, c.cert_name, c.issuing_organization
FROM students s
JOIN certifications c ON s.student_id = c.student_id
WHERE c.verification_status = 'Verified';

-- 4. Count skills per student (GROUP BY + ORDER BY)
SELECT s.name, COUNT(sk.skill_id) AS total_skills
FROM students s
LEFT JOIN skills sk ON s.student_id = sk.student_id
GROUP BY s.name
ORDER BY total_skills DESC;

-- 5. Students with more than one certification (Sub-query)
SELECT name FROM students
WHERE student_id IN (
  SELECT student_id FROM certifications
  GROUP BY student_id
  HAVING COUNT(*) > 1
);

-- 6. Update: mark a certification as verified
UPDATE certifications
SET verification_status = 'Verified'
WHERE cert_id = 2;

-- 7. Delete: remove a badge record
DELETE FROM badges WHERE badge_id = 3;
```

These queries collectively cover DDL (CREATE TABLE), DML (INSERT, UPDATE, DELETE), and DQL (SELECT with JOINs, WHERE, GROUP BY, ORDER BY, HAVING, and sub-queries). Together they showcase the expressive power of relational databases applied to the structured domain of student skill management.

6. Results and Output

The SkillBridge application was developed and tested with sample data. Four key screens were identified as the most representative outputs of the system, demonstrating the functional coverage of the platform from a user-interface perspective.

6.1 Dashboard Screen

The dashboard provides an overview of all key metrics for a logged-in student, including total skills recorded, number of certifications earned, workshops attended, and verified items. Visual metric cards and a Badges & Milestones panel offer an intuitive summary of the student's career readiness. The sidebar navigation gives access to all modules of the platform.

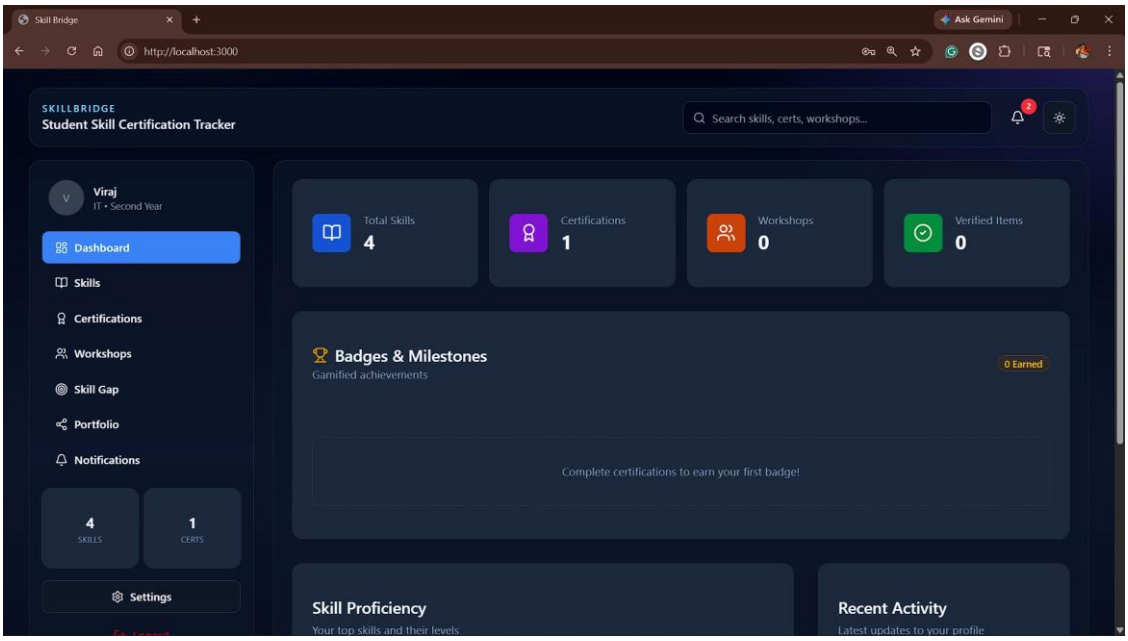

Figure 2: SkillBridge Dashboard — displaying student analytics and progress indicators

6.2 Student Skills Page

The Technical Skills page displays all skills entered by the student, categorized by domain (frontend, programming, database, design). Each skill card shows the proficiency level (Beginner / Intermediate / Advanced) and the last-updated date. Students can add new skills using the "Add New Skill" button, which triggers an insert operation into the skills table.

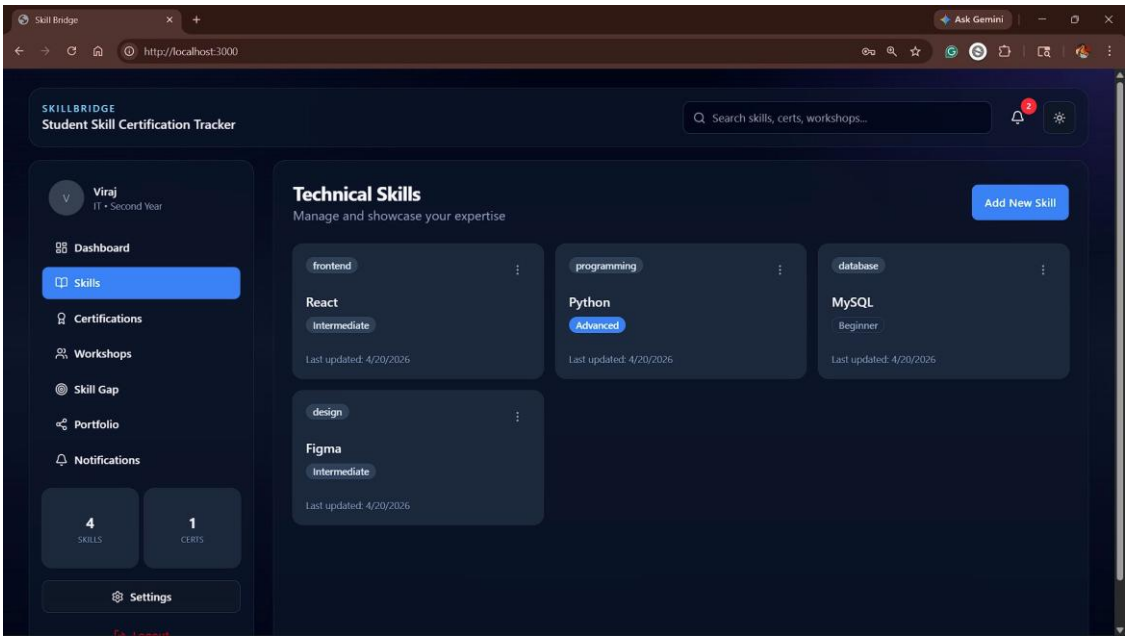

Figure 3: Technical Skills Page — managing and showcasing student expertise

6.3 Smart Certificate Upload Module

The Smart Certificate Upload module allows students to upload certificate files (PDF / PNG / JPG) along with metadata including the certificate title, issuing organization, and issue date. The AI-assisted module auto-extracts details from the uploaded file. Once submitted, the certificate enters a verification workflow where status is tracked as Pending, Verified, or Rejected — updating in real time via Supabase subscriptions.

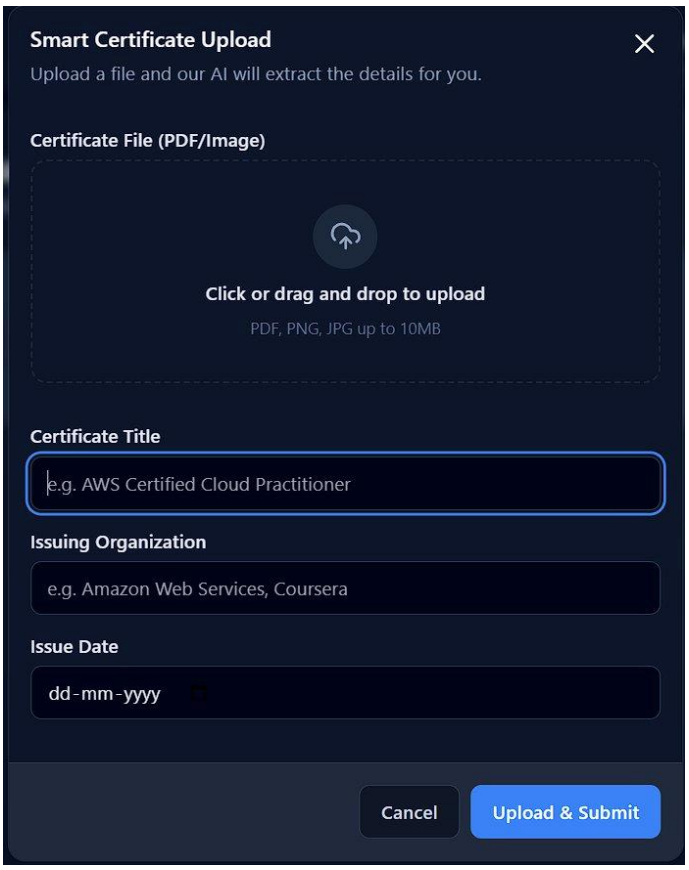

Figure 4: Smart Certificate Upload Module — with AI-assisted detail extraction

6.4 Public Portfolio / Student Profile Page

The Public Portfolio page displays the complete professional profile of a student in a shareable format. It shows a Skill Radar chart (visualizing proficiency across skills), an About Me section, Achievements, and a Workshops timeline. The page can be shared via a link or exported as a PDF resume. A "Verified Profile" badge is displayed for students whose credentials have been authenticated by the system.

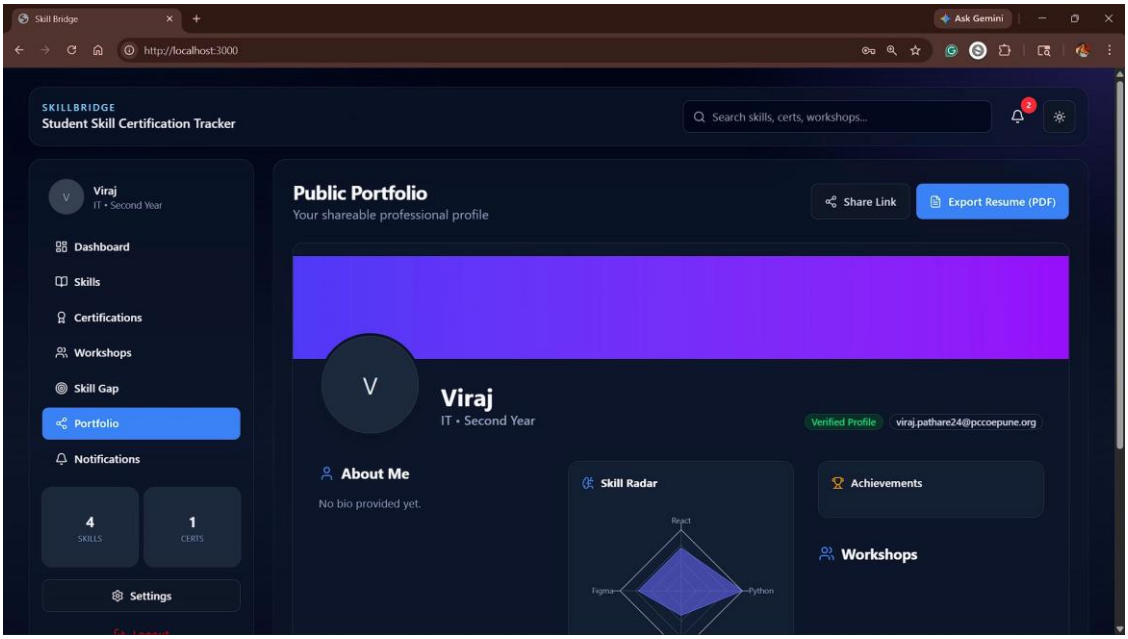

Figure 5: Public Portfolio Page — shareable professional profile with Skill Radar

7. Conclusion

The SkillBridge project successfully demonstrates the practical application of Database Management System concepts in solving a real-world academic challenge. By designing a normalized relational schema with five interconnected tables — Students, Skills, Certifications, Workshops, and Badges — the project showcases how structured data management can centralize and streamline student portfolio tracking. The use of primary keys, foreign keys, and relational joins ensures data integrity, consistency, and efficient retrieval across all modules of the application.

Through the development of this project, the team gained hands-on experience in ER diagram design, SQL query writing (DDL, DML, and DQL), database normalization, and integration with a modern backend-as-a-service platform (Supabase). The four key output screens — Dashboard, Technical Skills, Certificate Upload, and Public Portfolio — demonstrate the system's functional breadth and its potential for real-world deployment in an institutional setting.

The platform can be further enhanced in future iterations by integrating AI-based skill recommendations, automated certificate verification using OCR, real-time recruiter dashboards, and mobile application support. Overall, SkillBridge serves as a valuable tool for improving student career readiness, bridging the gap between academic achievement and industry expectations, and making student data verifiable, organized, and accessible in a structured manner.

8. References

- Supabase Official Documentation — https://supabase.com/docs
- PostgreSQL Documentation — https://www.postgresql.org/docs/
- W3Schools SQL Tutorial — https://www.w3schools.com/sql/
- React Official Documentation — https://react.dev/
- Tailwind CSS Documentation — https://tailwindcss.com/docs
- Ramez Elmasri & Shamkant B. Navathe, "Fundamentals of Database Systems", 7th Edition, Pearson Education.
- Abraham Silberschatz, Henry F. Korth & S. Sudarshan, "Database System Concepts", 7th Edition, McGraw-Hill.